

Entering the Movie Business: A Data-Driven Approach

Insights for new movie studio by Group 8



Group Members

Our team was composed of 4 data analysts who worked on the data. They include;

- Vihaan Sheth
- Kevin Kibet
- Vincent Ngochoch
- Chris Gitonga





Business Understanding



Problem Statement

The company wants to create a new movie studio to maximize profits and reduce risks.



Key Questions

Which genres have highest ratings and ROI? Which languages and directors succeed?



Goal

Ensure a strategic, well-informed entry into the movie sector.

Data Understanding

Data Sources

Box Office Mojo, IMDB, Rotten Tomato, The Movies Database, The Numbers Database.

Risk Minimization

Data guides smart investments by identifying high-performing genres, directors, and languages.

Business Strategy Support

Data-driven decisions align production with audience demand and commercial viability.



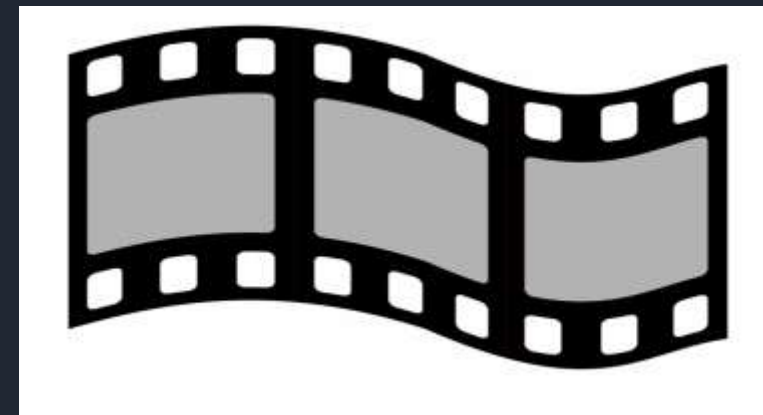
Business Objectives

Key Focus Areas

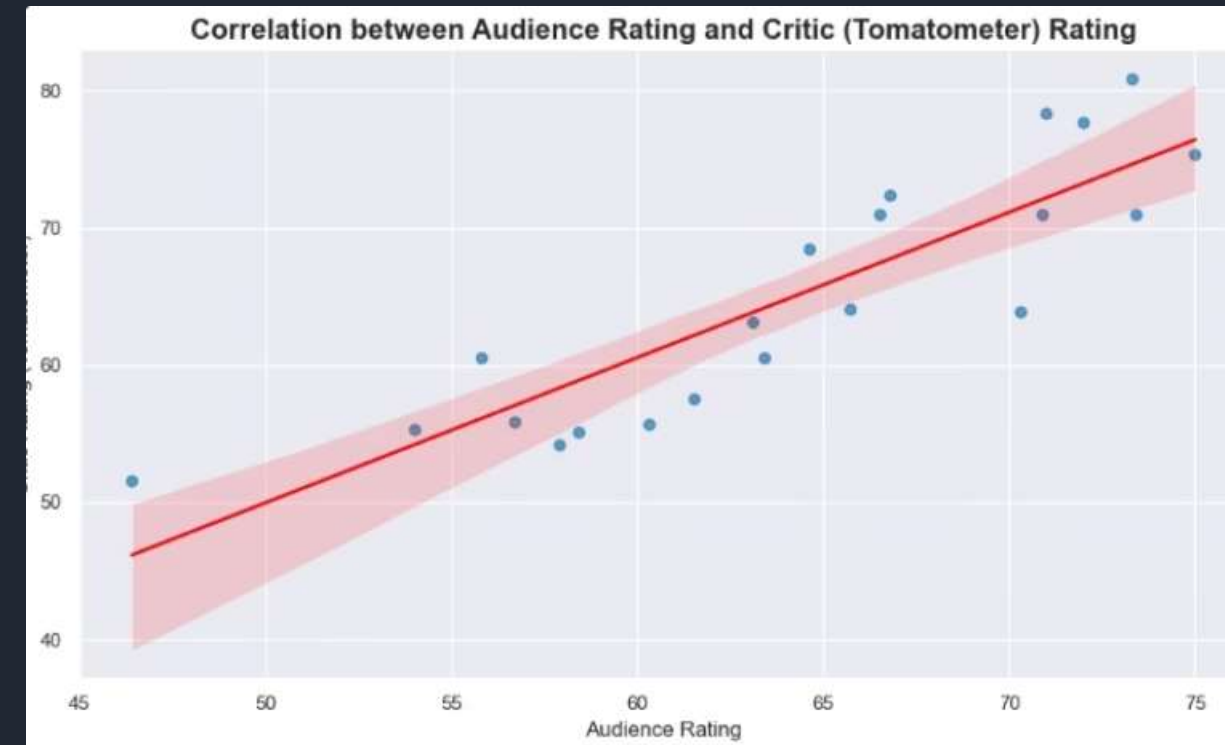
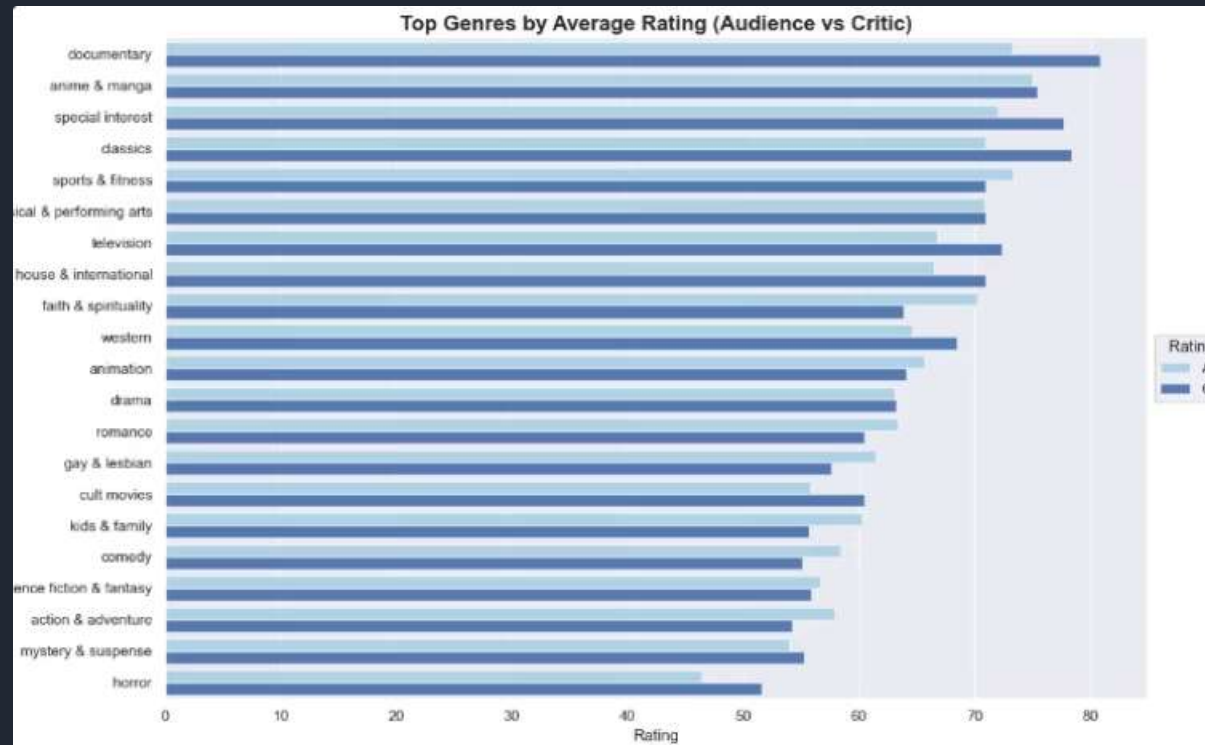
- Audience and critic rating trends by genre
- ROI trends by genre
- Language and popularity relationship
- Director worldwide gross earnings

Strategic Goals

- Maximize profitability and minimize financial risk
- Enhance creative confidence and audience satisfaction
- Identify global market opportunities by language
- Pinpoint successful directors for partnerships

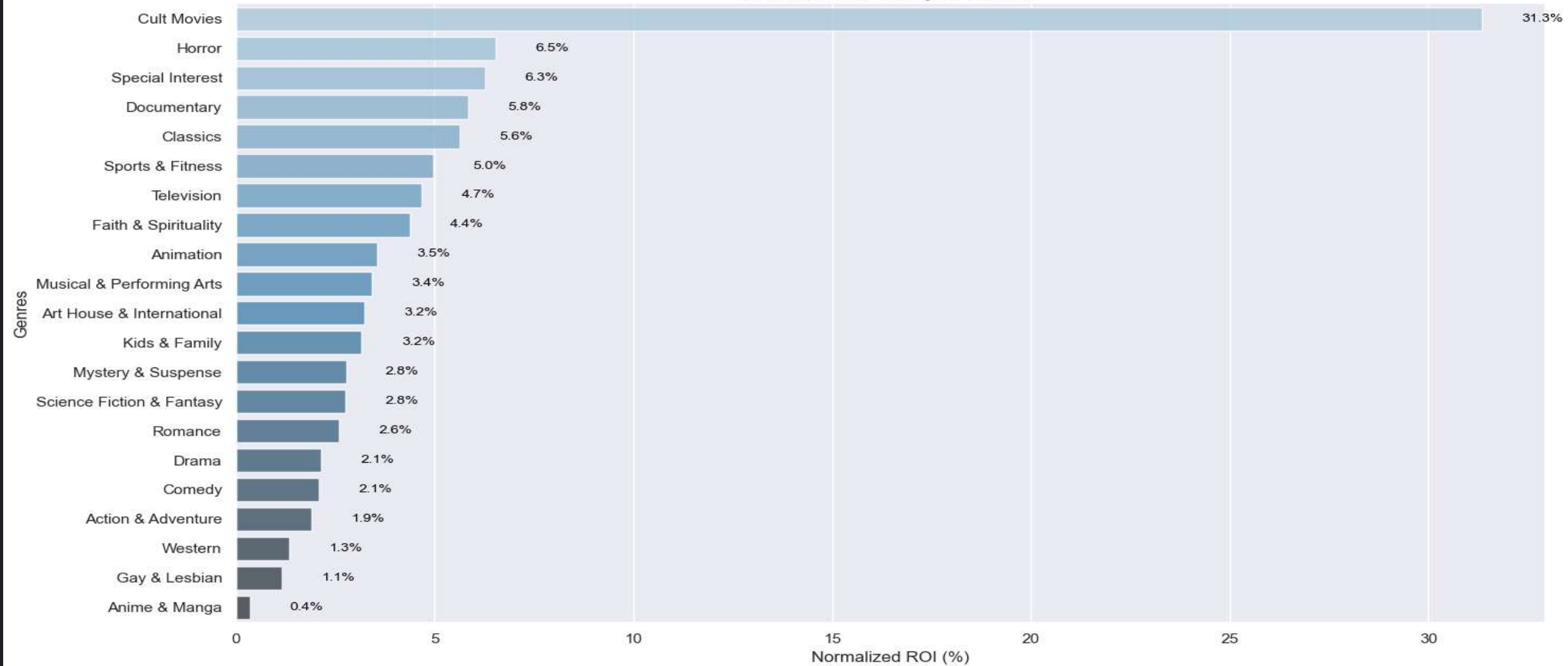


Audience and Critic Rating Trends by Genre



- **Documentaries** lead in critic and audience ratings.
- **Anime, Special Interest, Classics** also score highly with both groups.
- **Television** scores higher with audiences than critics.
- **Faith & Spirituality, Western** have lower ratings overall.
- **Sports & Fitness** appeals more to audiences than critics.
- The scatter plot indicates a positive correlation between audience preferences and critical acclaim, suggesting that popular genres tend to satisfy both audiences and critics without compromise x

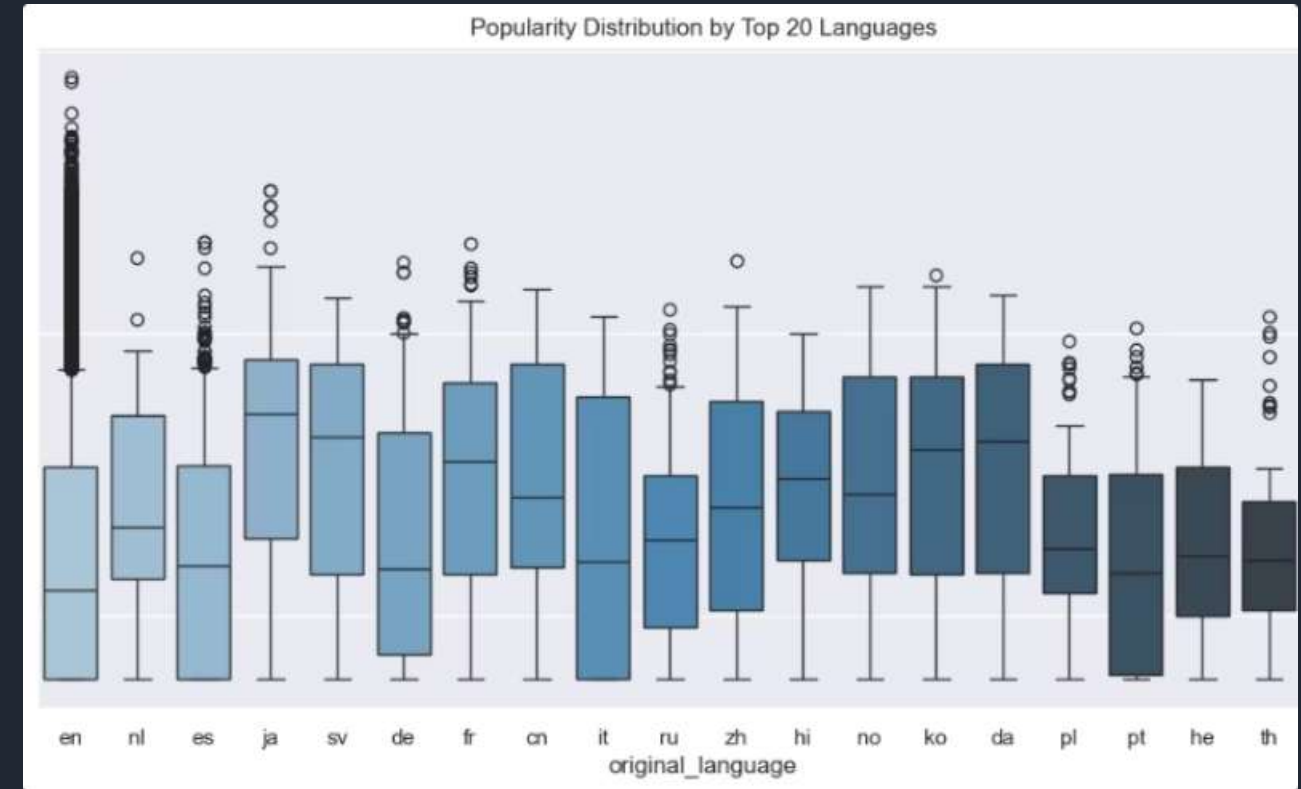
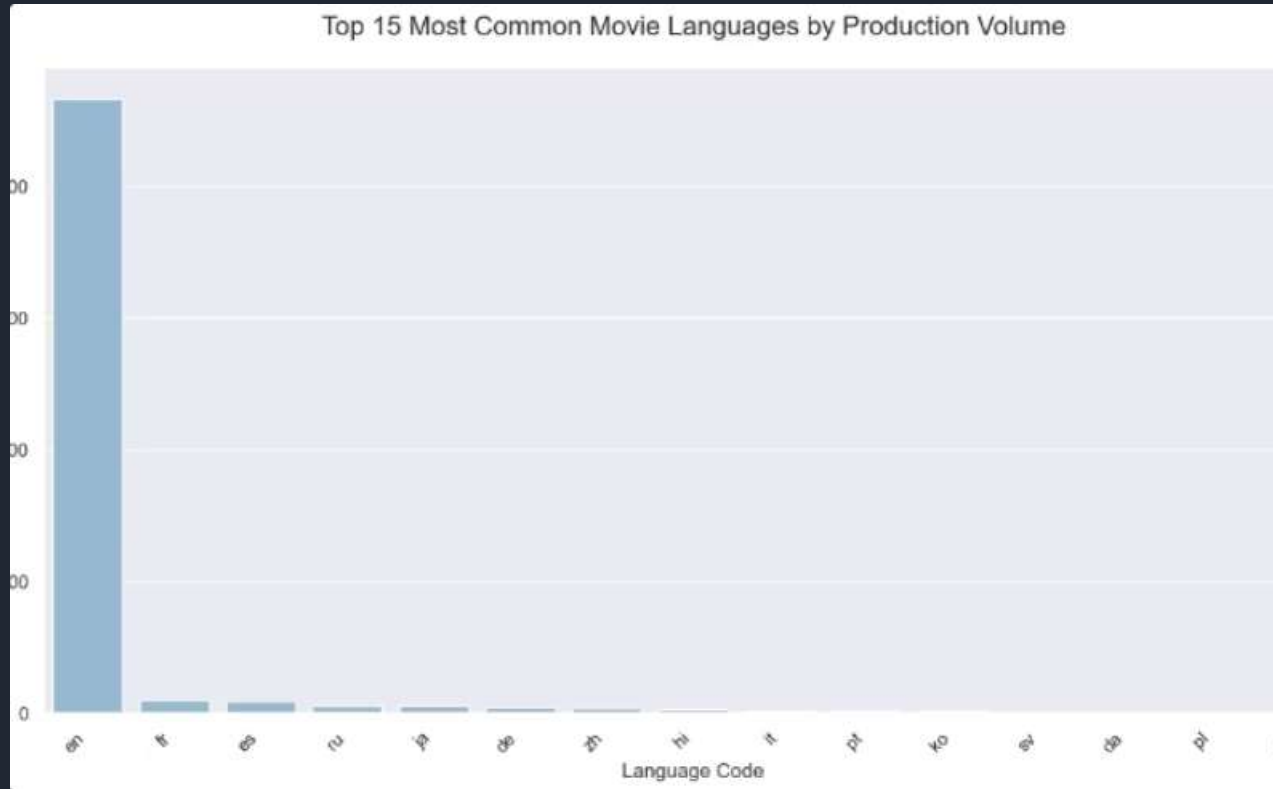
Normalized ROI per Genre



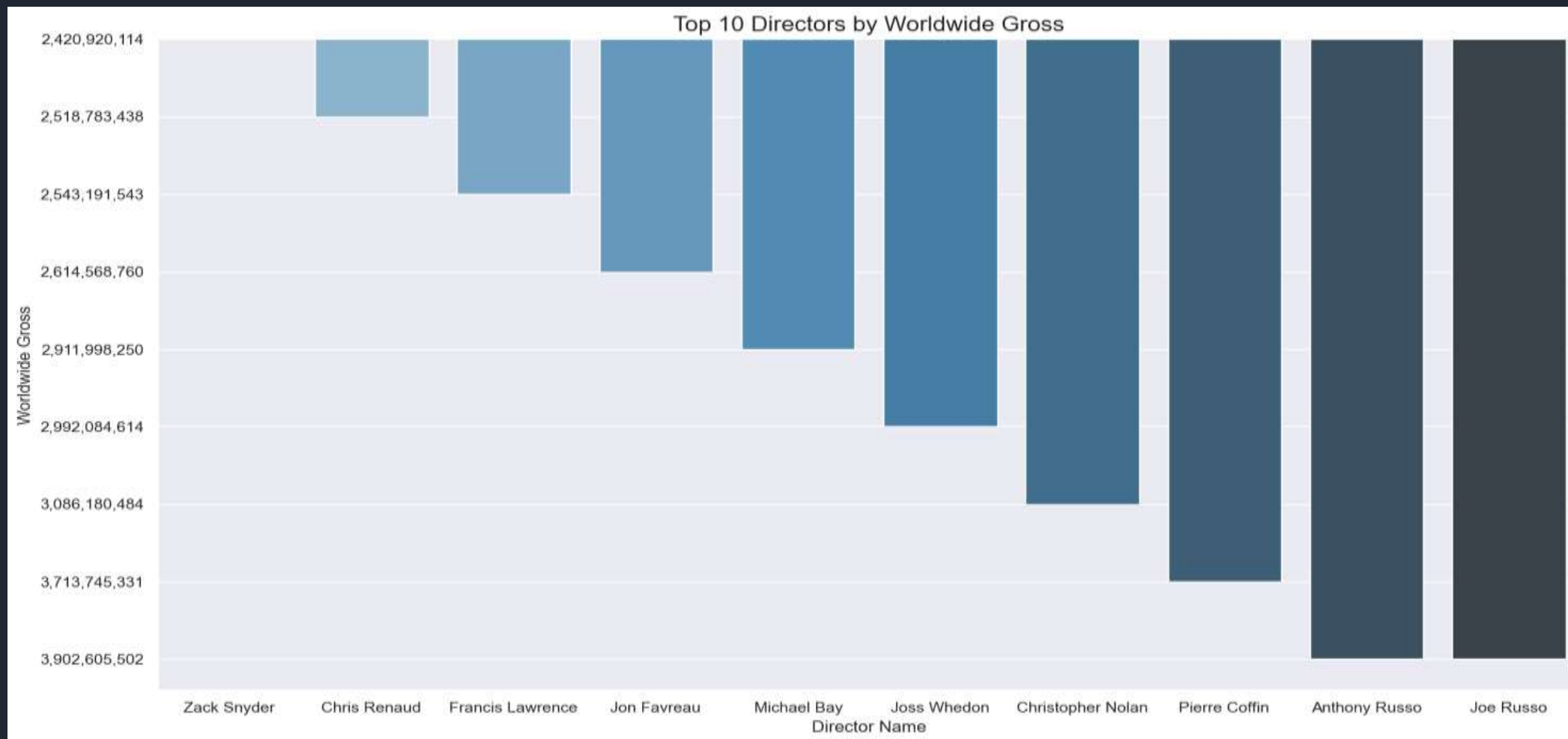
ROI Trends by Genre

- **Cult Movies** have the highest ROI (~31.3%), driven by niche loyal audiences.
- **Horror, Special Interest, Documentary** also show strong ROI with low budgets.
- **Action, Comedy, Drama** have lower ROI due to high production costs.
- **Western, Gay & Lesbian, Anime** show the lowest ROI, indicating limited reach.

Language and Popularity Relationship

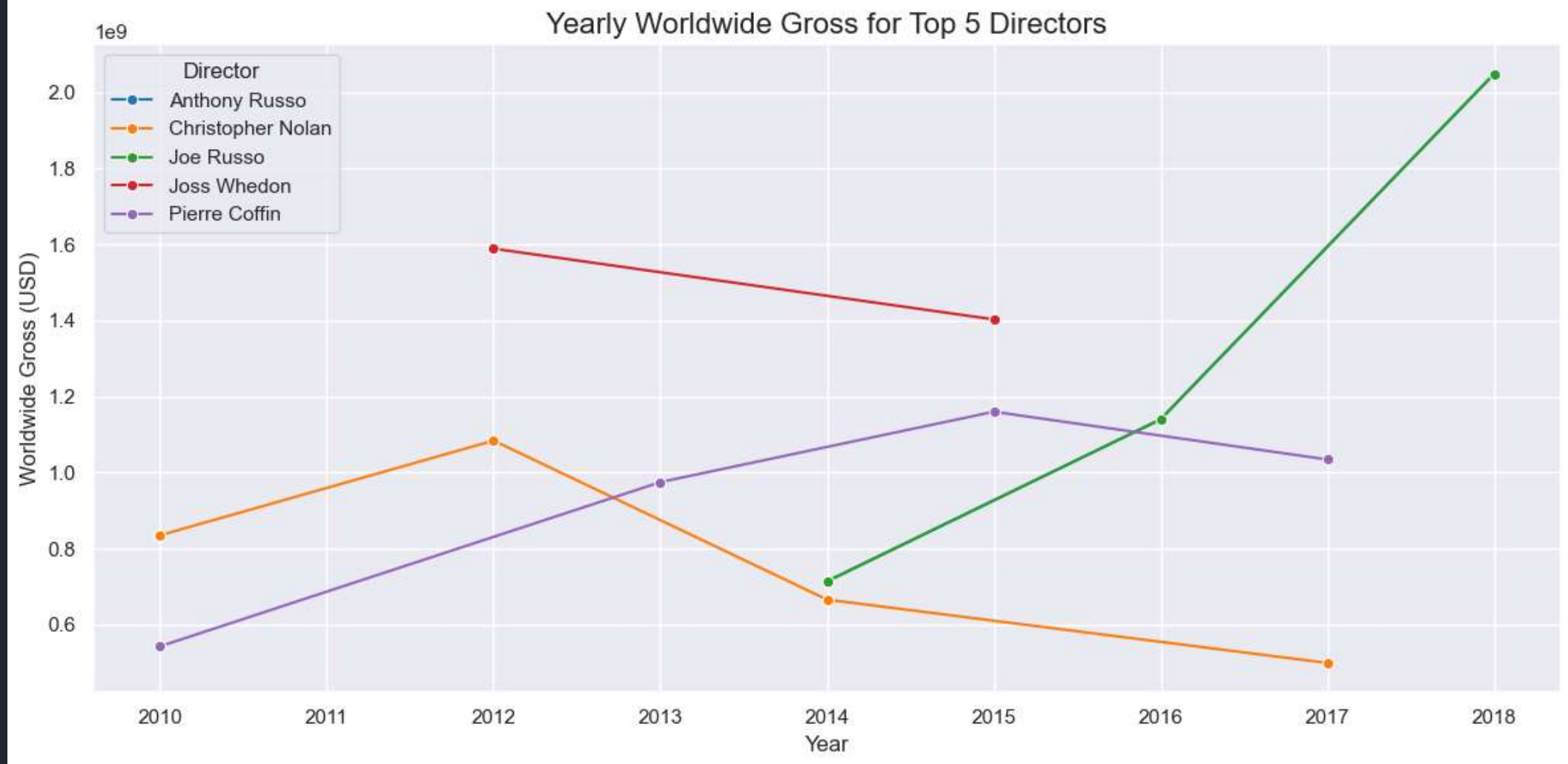


- **English** dominates with 20,000+ movies, followed by French and Spanish.
- English films show wide popularity with many top outliers.
- Japanese, Spanish, French, and Russian also produce highly popular films.
- Prioritizing English aligns with industry norms and maximizes reach.



Directors and Worldwide Gross Earnings

- **Russo Brothers** lead with nearly \$3.9B, driven by Marvel franchises.
- **Pierre Coffin** ranks high with animated hits like Despicable Me.
- **Christopher Nolan** shows success with original high-concept blockbusters.
- Action and franchise films dominate global box office success.
- There is a clear trend in that: franchise integration, visual spectacle, and genre appeal (especially action, animation, and superhero films) consistently lead to high global box office success including amazing directors



Director Performance Trends Over Time

- **Franchise Peaks:** Russo Brothers and Whedon spike during Marvel releases.
- **Animation Growth:** Coffin's steady rise shows reliable revenue from sequels.
- **Director Consistency:** Nolan's original films yield steady returns.
- **Strategic Timing:** Planned release cycles sustain momentum and earnings.

Business Recommendations

Focus on High-Impact Genres

Prioritize genres with critical and commercial strength like Documentaries and Cult Films.

Balance Ratings and Returns

Use low-budget, high-ROI genres like Horror for early projects.

Language Strategy

Start with English productions for maximum reach and plan multilingual releases.

Leverage Proven Directors

Collaborate with high-grossing directors and track emerging talent for long-term value.



Next Steps

- Commission pilot projects in high-ROI genres.
- Engage consultants or advisors from high-performing production backgrounds.
- Reassess strategy bi-annually with updated data and market trends

THANK YOU

Any questions ?

GROUP 8:

- I. Vihaan Sheth
- II. Kevin Kibe
- III. Vincent Ngochoch
- IV. Chris Gitonga

